Running GPUs in Kubernetes: From Setup to Scheduling and Sharing

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1. How to Enable GPUs in Kubernetes and Run a GPU Workload ?

In Kubernetes, GPUs are not available by default. A few components work together so that Kubernetes can discover GPUs and attach them to pods.

High-level GPU enablement flow

The following flow shows how Kubernetes detects and assigns GPUs:

```
GPU-enabled Pod
 ↓ (requests nvidia.com/gpu)

Kubelet (Kubernetes Cluster)
 ↓ (checks allocatable GPUs)

NVIDIA Device Plugin
 ↓ (discovers GPUs via driver)

NVIDIA Driver
```

```
 ↓
Physical GPU
```

———————— Pod Startup (runtime attachment path) ————————

```
Kubelet
 ↓ (invokes GPU-enabled runtime)

NVIDIA Container Runtime (Container Toolkit)
 ↓ (injects GPU devices & CUDA libs)

GPU-enabled Pod
```

By diagram we can understand, to enable GPUs in a cluster, three components must be configured on GPU nodes:

i. NVIDIA Driver :
This allows the operating system to communicate with the physical GPU hardware.

ii. NVIDIA Container Runtime (Container Toolkit) :
This injects GPU devices and CUDA libraries into containers at runtime.

iii. NVIDIA Device Plugin :
This discovers GPUs using the NVIDIA driver and advertises them to Kubernetes so pods can request them.

Step 1: Verify NVIDIA Driver on the GPU node

First, check whether the NVIDIA driver is installed on the GPU node:

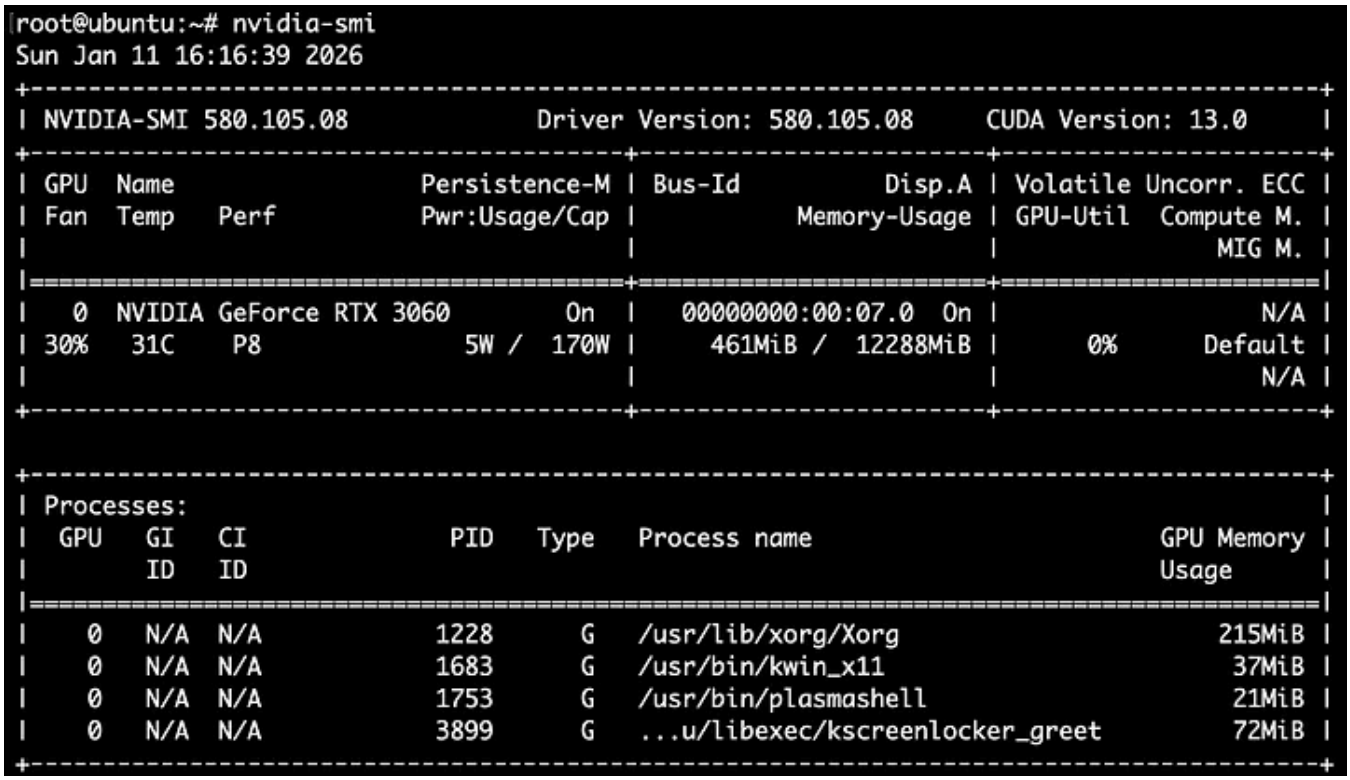
```
nvidia-smi
```

If the driver is installed correctly, this command will show GPU details.

If not, install the driver according to your operating system using the official documentation:

👉 https://docs.nvidia.com/datacenter/tesla/driver-installation-guide/index.html

After installation, we can verify it again using `nvidia-smi` .



Step 2: Install NVIDIA Container Runtime (Container Toolkit)

Next, we will install the NVIDIA Container Toolkit on the GPU node.

Follow the official documentation based on the operating system:

👉 https://docs.nvidia.com/datacenter/cloud-native/container-toolkit/latest/install-guide.html

After installation, we will run the following commands to attach the NVIDIA Container Runtime to the container runtime.

If using containerd (most Kubernetes clusters) :

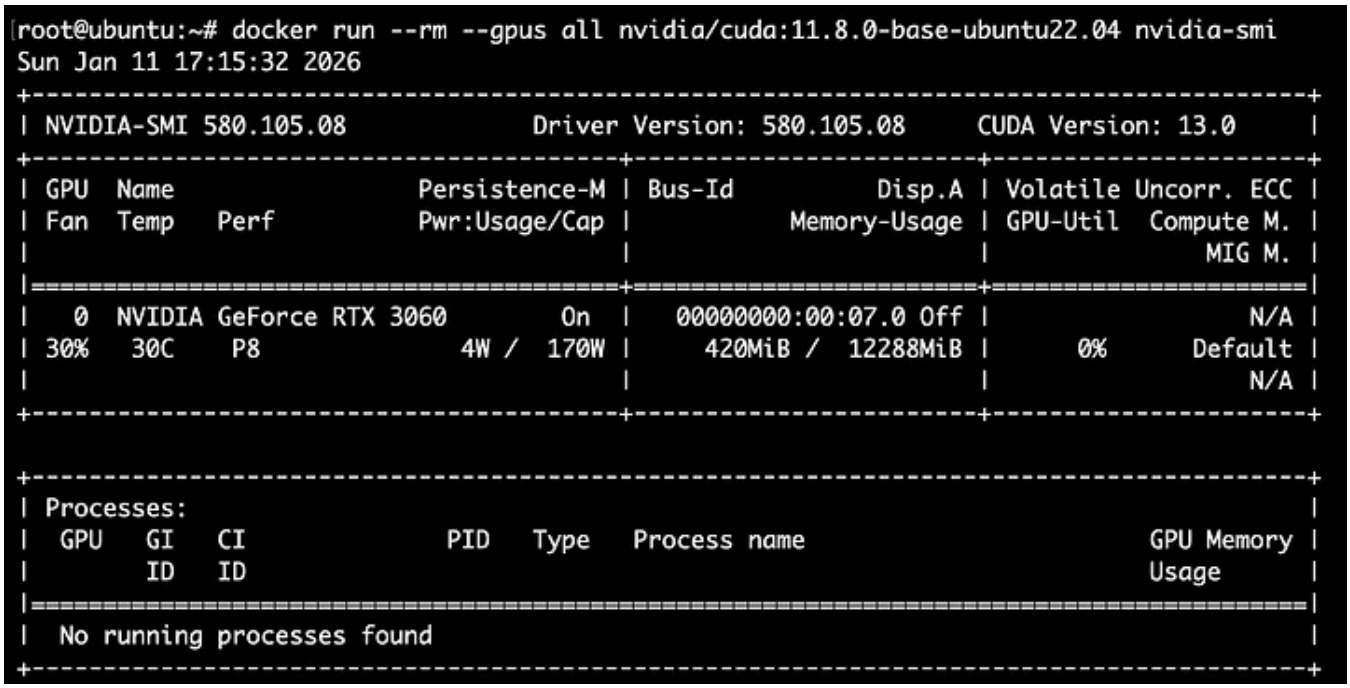
```
sudo nvidia-ctk runtime configure --runtime=containerd
sudo systemctl restart containerd
```

If using Docker :

```
sudo nvidia-ctk runtime configure --runtime=docker
sudo systemctl restart docker
```

After this step, containers can access GPUs when GPU resources are requested.

We can verify this in the following way if the NVIDIA Container Toolkit is configured with Docker :



Step 3: Install NVIDIA Device Plugin in the cluster

Finally, on the cluster, we can install the NVIDIA Device Plugin using the following command:

```
kubectl create -f https://raw.githubusercontent.com/NVIDIA/k8s-device-plugin/v0
```

After this step, the cluster is ready with GPU nodes.

We can verify this, and the GPU resources will now be visible on the nodes.

```
root@ubuntu:~# kubectl get pods -n kube-system | grep nvidia
nvidia-device-plugin-daemonset-rzltc      1/1     Running     0             2m8s
root@ubuntu:~# kubectl describe node ubuntu | grep -A 10 "Capacity:"
Capacity:
  cpu:                8
  ephemeral-storage:  76026616Ki
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             19321784Ki
  nvidia.com/gpu:     1
  pods:               110
```

Running a GPU workload in Kubernetes

To run a GPU-enabled pod, GPU resources must be explicitly requested:

```
resources:
  limits:
    nvidia.com/gpu: 1
```

Example: Simple GPU test pod

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Pod
metadata:
  name: gpu-pod-1
  labels:
    app: gpu-test
spec:
  restartPolicy: Never
  containers:
  - name: cuda-container
    image: nvidia/cuda:11.8.0-base-ubuntu22.04
    command:
    - /bin/bash
    - -c
    - |
      echo "GPU Pod 1 started"
      nvidia-smi
      echo "Running infinite loop to keep pod alive..."
      while true; do
        echo "GPU Pod 1 - $(date) - Running GPU workload simulation"
        nvidia-smi --query-gpu=utilization.gpu,memory.used --format=csv
        sleep 10
      done
    resources:
```

```
        limits:
          nvidia.com/gpu: 1
EOF
```

And now we can verify it.

```
root@ubuntu:~# kubectl get pods gpu-pod-1
NAME        READY   STATUS    RESTARTS   AGE
gpu-pod-1   1/1     Running   0          66s
root@ubuntu:~# kubectl logs gpu-pod-1 --tail=20
utilization.gpu [%], memory.used [MiB]
0 %, 420 MiB
GPU Pod 1 - Sun Jan 11 17:55:04 UTC 2026 - Running GPU workload simulation
utilization.gpu [%], memory.used [MiB]
0 %, 420 MiB
GPU Pod 1 - Sun Jan 11 17:55:04 UTC 2026 - Running GPU workload simulation
utilization.gpu [%], memory.used [MiB]
0 %, 420 MiB
GPU Pod 1 - Sun Jan 11 17:55:04 UTC 2026 - Running GPU workload simulation
utilization.gpu [%], memory.used [MiB]
0 %, 420 MiB
GPU Pod 1 - Sun Jan 11 17:55:04 UTC 2026 - Running GPU workload simulation
utilization.gpu [%], memory.used [MiB]
0 %, 420 MiB
GPU Pod 1 - Sun Jan 11 17:55:04 UTC 2026 - Running GPU workload simulation
utilization.gpu [%], memory.used [MiB]
0 %, 420 MiB
GPU Pod 1 - Sun Jan 11 17:55:04 UTC 2026 - Running GPU workload simulation
utilization.gpu [%], memory.used [MiB]
0 %, 420 MiB
```

2. What happens when multiple pods request more GPUs than available ?

Since we already deployed one GPU pod and verified via resources that we have only one GPU node, now we want to check what happens if we deploy another pod that also requires a GPU.

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: Pod
metadata:
  name: gpu-pod-2
  labels:
    app: gpu-test
spec:
  restartPolicy: Never
  containers:
```

```
    - name: cuda-container
      image: nvidia/cuda:11.8.0-base-ubuntu22.04
      command:
      - /bin/bash
      - -c
      - |
        echo "GPU Pod 2 started"
        nvidia-smi
        echo "Running infinite loop to keep pod alive..."
        while true; do
          echo "GPU Pod 2 - $(date) - Running GPU workload simulation"
          nvidia-smi --query-gpu=utilization.gpu,memory.used --format=csv
          sleep 10
        done
      resources:
        limits:
          nvidia.com/gpu: 1
EOF
```

Now, if we check the pod status, the second pod will remain in the **Pending** state because there are no available GPU resources. We can verify this by describing the pod.

```
root@ubuntu:~# kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
gpu-pod-1   1/1     Running   0          107s
gpu-pod-2   0/1     Pending   0          4s
```

```
root@ubuntu:~# kubectl describe pod gpu-pod-2 | grep -A 10 "Events:"
Events:
  Type     Reason            Age   From               Message
  ----     ------            ----  ----               -------
  Warning  FailedScheduling  36s   default-scheduler  0/1 nodes are available: 1 Insufficient nvidia.com/gpu. no new claims to deallocate, preemption: 0/1 nodes are available: 1 No pre
emption victims found for incoming pod.
```

3. What is GPU time slicing, and how can we configure it ?

GPU Time Slicing is a way to share a single GPU among multiple processes **without physically partitioning or splitting the GPU**. Instead, each process gets access to the GPU for a short time slice, one after another.

When we enable time slicing, Kubernetes exposes **virtual (fake) GPU resources**, allowing multiple pods to request GPUs even though there is only one physical GPU. These virtual GPUs are available to processes as long as the GPU is not fully saturated by another workload.

```
### Before Time-Slicing
┌──────────────────────┐
│  Physical GPU 0      │  → Can run 1 pod
└──────────────────────┘

### After Time-Slicing (replicas: 4)
┌──────────────────────┐
│  Physical GPU 0      │
├──────────────────────┤
│  Virtual GPU 0       │  → Pod 1
│  Virtual GPU 1       │  → Pod 2
│  Virtual GPU 2       │  → Pod 3
│  Virtual GPU 3       │  → Pod 4
└──────────────────────┘
```

Steps to configure GPU time slicing

1. Create a ConfigMap for time slicing

```
cat <<EOF | kubectl apply -f -
apiVersion: v1
kind: ConfigMap
metadata:
  name: time-slicing-config
  namespace: kube-system
data:
  any: |-
    version: v1
    flags:
      migStrategy: none
    sharing:
      timeSlicing:
        resources:
        - name: nvidia.com/gpu
```

```
            replicas: 4
    EOF
```

This configuration creates **4 virtual GPUs** from a single physical GPU using time slicing.

2. Patch the NVIDIA Device Plugin DaemonSet to use this ConfigMap

```
kubectl patch daemonset nvidia-device-plugin-daemonset -n kube-system --type='j
  {
    "op": "replace",
    "path": "/spec/template/spec/volumes",
    "value": [
      {
        "name": "device-plugin",
        "hostPath": {
          "path": "/var/lib/kubelet/device-plugins"
        }
      },
      {
        "name": "time-slicing-config",
        "configMap": {
          "name": "time-slicing-config-all"
        }
      }
    ]
  }
]'
```

Now, after the NVIDIA Device Plugin pod restarts, we can verify the GPU resources in the following way.

```
root@ubuntu:~# kubectl describe node ubuntu | grep -A 10 "Capacity:"
Capacity:
  cpu:                8
  ephemeral-storage:  76026616Ki
  hugepages-1Gi:      0
  hugepages-2Mi:      0
  memory:             19321784Ki
  nvidia.com/gpu:     4
  pods:               110
```

Now we can also verify those pods.

```
root@ubuntu:~# kubectl get pods
NAME        READY   STATUS    RESTARTS   AGE
gpu-pod-1   1/1     Running   0          14m
gpu-pod-2   1/1     Running   0          12m
root@ubuntu:~# kubectl get pod gpu-pod-1 -o jsonpath='{.spec.containers[0].resources}' | jq .
{
  "limits": {
    "nvidia.com/gpu": "1"
  },
  "requests": {
    "nvidia.com/gpu": "1"
  }
}
root@ubuntu:~# kubectl get pod gpu-pod-2 -o jsonpath='{.spec.containers[0].resources}' | jq .
{
  "limits": {
    "nvidia.com/gpu": "1"
  },
  "requests": {
    "nvidia.com/gpu": "1"
  }
}
```

As expected, **both GPU pods will now be in the Running state**, and both will be utilizing the same physical GPU through time slicing.

4. What is the NVIDIA GPU Operator, and how does it simplify GPU setup and management ?

Instead of manually managing all the steps we covered earlier, GPU setup and management can be handled much more easily using the **NVIDIA GPU Operator**. The GPU Operator automates the entire GPU lifecycle in Kubernetes. It not only handles the steps mentioned above but also provides additional functionality. In short, it manages everything in a much cleaner and more scalable way.

> *What the NVIDIA GPU Operator provides by default*

- **GPU Driver Management**
  Automatically installs and manages NVIDIA GPU drivers across all GPU nodes (especially useful when the cluster has multiple GPU nodes).
- **NVIDIA Container Runtime**
  Enables containers to access GPUs using the NVIDIA container runtime.
- **NVIDIA Device Plugin**
  Exposes GPU resources to Kubernetes so pods can request GPUs.

- **GPU Feature Discovery**
  Labels nodes with GPU details such as model, memory, and compute capabilities.
- **DCGM Monitoring**
  Collects GPU metrics like utilization, memory usage, temperature, and power.
- **DCGM Exporter**
  Exports GPU metrics to Prometheus for monitoring and alerting.
- **CUDA Toolkit (Optional)**
  Provides CUDA libraries inside containers when required by workloads.
- **MIG Management (If Supported)**
  Automatically configures and manages MIG instances on supported GPUs.
- **Node Validation**
  Runs health checks to ensure GPU nodes are correctly configured.
- **Upgrade & Lifecycle Management**
  Handles driver and component upgrades without manual intervention.

For a deeper explanation, refer to the official documentation:

👉 https://docs.nvidia.com/datacenter/cloud-native/gpu-operator/latest/overview.html

Steps to install the NVIDIA GPU Operator

1. Add the NVIDIA Helm repository

```
helm repo add nvidia https://helm.ngc.nvidia.com/nvidia && helm repo update
```

2. Install the GPU Operator

Install the operator with the default configuration:

```
helm install --wait --generate-name \
  -n gpu-operator --create-namespace \
```

```
    nvidia/gpu-operator \
    --version=v25.10.1
```

3. Enable Time Slicing

To enable GPU time slicing, there is only need to create the time-slicing configuration file and pass it during the Helm install or upgrade command. The GPU Operator will handle the rest.

If GPU Operator is already installed:

```
helm upgrade gpu-operator nvidia/gpu-operator \
  -n gpu-operator \
  --set-file devicePlugin.config=time-slicing-config.yaml
```

After installation, it can be verified in the same way as done in the previous steps.

Big idea:
Before the NVIDIA GPU Operator, setting up GPUs in Kubernetes was chaotic. Drivers, container runtimes, device plugins, and monitoring had to be installed and managed manually.

GPU Operator = one Helm install, and NVIDIA handles everything.

5. What are other concurrency mechanisms in NVIDIA GPUs ?

Multi-Instance GPU (MIG)

On newer NVIDIA GPU models, it is possible to achieve **complete isolation at the physical level** using Multi-Instance GPU (MIG).

Unlike GPU time slicing, where multiple processes share the same GPU memory and compute resources without actual partitioning, MIG **physically partitions a single GPU** into multiple isolated instances. Each instance has its own dedicated memory, compute, and cache resources.

MIG is available only on **newer NVIDIA GPU models** and is useful when strong isolation and predictable performance are required.

Gpu | Mlops | Ai Infrastructure | Time Slicing | Gpu Sharing

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